

Note: I). Answer all the questions. II). Each correct answer carries 2 marks

(10X2=20)

1. Write the definitions of a). Histology b). Osteology. (2m)

A: - a). Histology – study of tissues b). Osteology – study of bones.

2. List out cytoplasmic organelles.(2m)

A:- Cytoplasmic organelles are Endoplasmic reticulum, Golgi apparatus, Mitochondria, Lysosomes, Ribosomes, Centrosomes etc.

3. Define respiration .(2m)

A:- :- Respiration is the process of gaseous exchange between blood and external environment in lungs and blood and body tissues.

4. Write the arrangement of permanent teeth. (2m)

A: - I C P M 2.1 2.3 I – incisor, C – canine, P – Premolar, M – Molar teeth
I C P M 2.1 2.3

5. What are the layers of heart? .(2m)

A:- Outer layer:- pericardium, middle layer – myocardium, inner most layer – endocardium.

6. Mention main functions of blood .(2m)

A:- - 1). Transport of oxygen from lungs to tissues and carbon dioxide from tissues to lungs. 2). Transport of end products of digestion absorbed from intestines to cells for utilization. 3). Carriage of essential chemicals like hormones and vitamins. 4). Transport of waste products.

7. What is nervous tissue?.(2m)

A: - Nervous tissue is excitable type tissue receiving and transmitting messages. It is composed of neurons and neuroglia.

8. Mention the parts of urinary system.(2m)

A: -. Kidneys, ureters, urinary bladder, urino genital tract / urethra.

9. Define a). Hormone. b). Endocrine gland. (2m)

A: -. a). **Hormone:-** Hormones are chemical substances produced by endocrine glands and are poured into blood stream. They reach the target cells through blood.

b). **Endocrine gland:-** Endocrine glands are ductless glands whose secretions pour into blood directly.

10. Name the parts of spermatozoa.(2m)

A:-. The parts of spermatozoa are head, neck, body and tail.

SECTION-B

Note: i). Answer any five questions ii). Each correct answer carries 6 marks. 5X6=30

11. Classify tissues.

A:- **Classification of tissues:-** Tissues are mainly classified into four types. They are 1). Epithelial tissue 2). Connective tissue 3). Muscular tissue 4). Nervous tissue.(2m)

1). Epithelial tissue:- It gives covering to other tissues by forming epithelial membrane. They are sub divided into simple epithelium and compound epithelium. (1m)

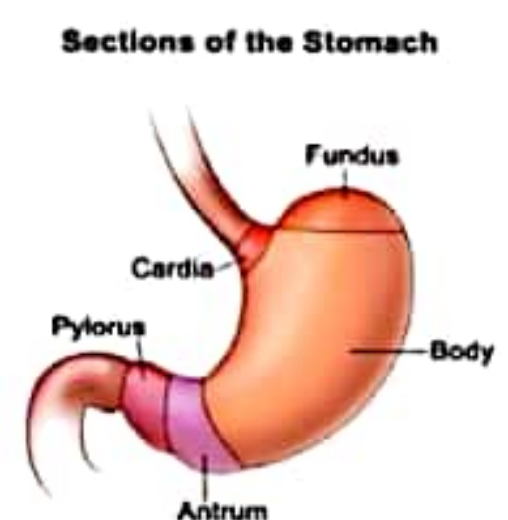
2). Connective tissue:- It is developed from mesoderm. It serves the function of binding two tissues(1m)

3). Muscular tissue:- It is the type of tissue having contractile ability on excitation. It is divided into voluntary and involuntary muscles. (1m)

4). Nervous tissue:- Nervous tissue is constitutes nervous system. Nervous tissue is excitable type tissue receiving and transmitting messages. (1m).

12. Write the anatomy of stomach. Draw the diagram and label.(3+3)

A:- **The anatomy of stomach:-** Stomach is the most dilatable part of digestive tract. It is J shaped. It is situated between the end of the oesophagus and beginning of the small intestine. It lies below the diaphragm in the abdominal cavity. Its major part is to the left of the mid line. It distends when it is filled with food. Average capacity of stomach is 1.5ltrs. in adult. Stomach has two surfaces, two curvatures two ends, three parts and two sphincters. They are as follows.



The two surfaces of stomach are:- 1). Anterior surface 2). Posterior surface.

Two curvatures are :- 1). Lesser curvature 2). Greater curvature.

Three parts of stomach are:- 1). Funds(upper portion)-above the cardiac sphincter 2). Body(middle portion 3). Pylorus(lower portion). There are two sphincters. A). Cardiac sphincter b). Pyloric sphincter.

Histologically there are four coats in the stomach wall a). Outer serous coat b). Muscular coat c). Sub mucous coat d). Mucous coat.

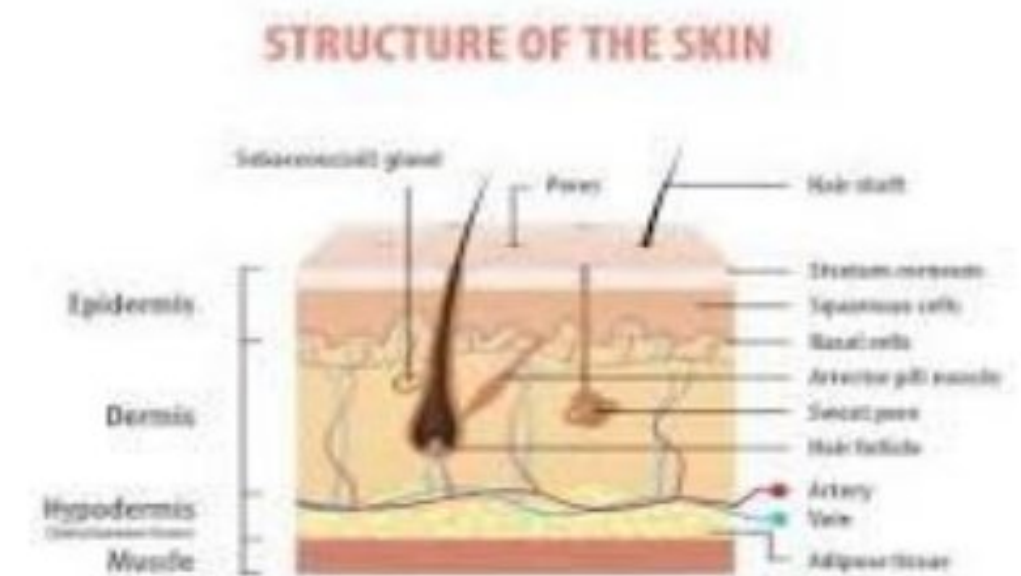
13. Describe the structure of skin. Draw diagram.(3+3)

A:- The skin is the body's largest and primary protective organ covering its entire external surface and serving as first order. It consists of two layers. They are 1). Epidermis – outer layer 2). Dermis – inner layer.

Epidermis is made of stratified epithelium. Layers of epidermis are 1). Stratum corneum. 2). Stratum lucidum 3). Stratum granulosum 4). Stratum germinatum.

Dermis – Dermis is inner layer of skin. It contains melanophore cells containing melanin pigment, arterial and venous capillaries and also sensory nerve endings, sweat glands and sebaceous glands, roots of hairs, erector pili muscles etc.,

Various sensory nerve endings of skin are encapsulated in connective tissue and are responsible for receiving various sensations.



14. Explain the anatomy of heart with figure.(3+3)

A:- **The anatomy heart:-** Heart lies on the left upper part of thoracic cavity. It lies between the two lungs under sternum it is broad above and conical below. It is in the size of a closed fist. It weights about 300gm.

Histology of heart:- Heart consists of three layers.

Outer layer - pericardium, middle layer – myocardium, inner most layer – endocardium.

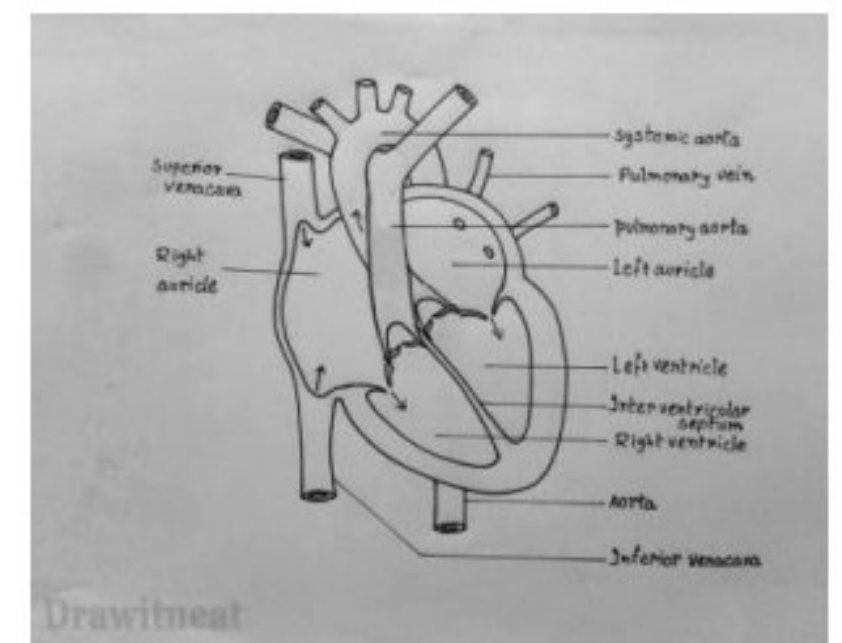
Pericardium forms bag like structure between visceral and parietal layers containing pericardial fluid.

Chambers of heart:- Heart has four chambers two of them of are upper chambers called atria or auricles. Lower two chambers are called ventricles. The two atria are separated by inter atrial septum. The two ventricles are separated by inter ventricular septum.

Atria are filling chambers and ventricles pumping chambers. Compared to atria ventricles are thicker since they are pumping chambers. Of the two ventricles wall of the left ventricle is three times thicker then the right ventricle.

Valves of heart:- Opening between right atrium and right ventricle is guarded by tricuspid valve. Opening between left atrium and left ventricle is guarded by bicuspid or mitral valve.

Blood vessels attached to heart:- 1). Superior and inferior vena cavae 2). Pulmonary artery 3). Pulmonary veins 4). Aorta.



15. Write about mechanism of respiration.

A:- :- There are two stages in respiration. They are 1). Inspiration 2). Expiration. Inspiration means breathing in. Expiration means exhalation of gases. Inspiration and expiration are brought about by the respiratory centres present in medulla oblongata. They are inspiratory and expiratory centres.(2m)

Mechanism of inspiration:- It is an active process. Inspiratory centres discharges the impulses rhythmically. They are carried by phrenic and intercostal nerves. They result in contraction of respiratory muscles – diaphragm and intercostals muscles. (2m)

Mechanism of expiration:- It is a passive process. Expiratory centres inhibit inhibitory centres and result in relaxation of diaphragm and intercostal muscles. It causes reduction in the size of thoracic cavity(2m)

16. Classify joints. What are different synovial joints?

A:- Joints are classified into three major structural forms and on their mobility namely a) Fibrous joints, b) Cartilaginous joints and c) Synovial joints.

A) **Fibrous joints(Immovable joints):** The fibrous joints do not allow any movement. These are composed of dense irregular connective tissue. Ex:- Sutures or Joints of the flat skull bones

B) **Cartilaginous joints(slightly movable joints):** These are slightly movable joints in which the joint surfaces are separated by some intervening substance and slight movement only is possible. Ex:- The pubic symphysis, inter vertebral joints.

C) **Synovial joints(Freely movable joints):** These are characterized by the presence of a fluid filled synovial cavity between the articulating surfaces of the two bones. The articular surfaces of the bones are covered by hyaline cartilage. Ex:- Knee joint, elbow joint.(4m)

Types of Synovial Joints: 1). Ball and socket joint 2). Hinge joint 3). Pivot joint 4). Gliding joint 5). Condylloid joint 6). Saddle joint(2m)

17. Write the functions of kidneys. (6m)

A:- 1). Excretion of waste products of cellular metabolism

2). Maintenance of PH.

3). Maintenance of water balance.

4). Maintenance of acid base equilibrium.

5). Maintenance of osmotic pressure in blood and tissues.

6). Elimination of drugs and toxic from body.

7). Regulation of blood pressure.

8). Manufacture of new substances.

9). Regulation of erythropoiesis.

10). Playing important role in vitamin-D metabolism.

11). Destruction of insulin and glucagon hormones takes place in kidneys.

12). Selective reabsorption.

13). Kidneys act as receptor organ.

18. List out female sex hormones. Add a note on them.

A:- **Femal sex hormones:-** They are oestrogens, progesterone, relaxin and androgens. (2m)

Oestrogens:- It is produced by graafian follicle of ovaries. In pregnancy, placenta produces large amounts of oestrone. (1m)

Progesterone:- It is secreted by corpus luteum of ovaries and placenta. (1m)

Functions:- oestrogens and progesterone are responsible for growth of female sex organs, secondary sexual characters. It plays role in menstrual cycle and in pregnancy.

Relaxin:- It is the hormone present in pregnant mammalian ovary, placenta and uterus. It is useful in affecting parturition. (1m)

Androgens:- Small amounts of testosterone present in females are responsible for development of libido. (1m)

SECTION-A

Max. Marks=50

(10X2=20)

Note: Answer all the questions. Each correct answer carries 2 marks**1. Name the chambers of heart ?(2m)**

A:- 1). Right atrium 2). Right ventricle 3). Left atrium 4). Left ventricle.

2. Define physiology. (2m)

A: - Physiology is the science of study of normal functions of body.

3. Name the stages of respiration.(2m)

A:- 1). Ventilation 2). Intrapulmonary gas mixing 3). Diffusion 4). Perfusion.

4. Write the definition of synapse ?(2m)

A:- Synapse is the junction where one neuron ends and another neuron begins.

5. Define blood(2m)

A:- Blood is defined as specialised fluid connective tissue of body containing blood cells suspended in plasma..

6. What are functions of ear ? (2m)

A: -The functions of ear are hearing and equilibrium of body.

7. Write any four functions of saliva. (2m)

A:- 1.It helps to keep the mouth moist and facilitates speech 2.It helps in mastication and formation of bolus 3.It dilutes hot and pungent substances does preventing injury to the mouth 4.It acts as solvent for soluble substances 5. It helps in lubrication and cleaning of mouth.6. Ptyalin present in saliva converts starch into maltose.7. Saliva helps in excretion of heavy metals urea and certain drugs 8.It helps in maintenance of water balance by nervous mechanism.9. It acts as buffer.

8. What is dialysis ? (2m)

A:- Dialysis is the process of diffusing the incompatible substances out of blood by means of artificial device .Arterial blood passed through this device and after filtration blood is returned back into the body through the vein.

9. Define a). Hormone. b). Endocrine gland. (2m)A: - **a). Hormone:-** Hormone is a chemical substance produced by endocrine gland and is poured into blood stream. It reach the target cells through blood.**b). Endocrine gland:-** Endocrine gland is a ductless gland, whose secretions pour into blood directly.**10. Mention the parts of uterus .(2m)**

A:- The parts of uterus are fundus, body and cervix.

SECTION-B

Note: i). Answer any five questions ii). Each correct answer carries 6 marks. 5X6=30**11. Classify tissues.(6m)**A:- **Classification of tissues:-** Tissues are mainly classified into four types. They are 1). Epithelial tissue 2). Connective tissue 3). Muscular tissue 4). Nervous tissue.(2m)**1). Epithelial tissue:-** It gives covering to other tissues by forming epithelial membrane. They are sub divided into simple epithelium and compound epithelium. (1m)**2). Connective tissue:-** It is developed from mesoderm. It serves the function of binding two tissues(1m)**3). Muscular tissue:-** It is the type of tissue having contractile ability on excitation. It is divided into voluntary and involuntary muscles. (1m)**4). Nervous tissue:-** Nervous tissue is constitutes nervous system. Nervous tissue is excitable type tissue receiving and transmitting messages. (1m).**12. Write the anatomy of respiratory system.(6m)**

A:- Respiratory system is consisting of Nose, Pharynx, Larynx, Trachea, Bronchi, Bronchioles, Alveolar ducts and alveoli.

Nose: It is the part of respiratory system through which air is inhaled in and exhaled out. It consists of external nose, nasal cavity, anterior nares, posterior nares, roof and floor.

-: 2 :-

Pharynx: It lies between nasal cavity and larynx. **Larynx :** It lies below the pharynx and above trachea. it is divided into 1). Nasopharynx 2). Oropharynx 3). Laryngopharynx. **Trachea:** It is also called as wind pipe it is a cylindrical tube it is 11cm. in length. It is made of 16 to 20 C shaped incomplete cartilages.

Bronchi: Trachea divide into right and left bronchi.

Bronchioles: Bronchioles are the finest branches of bronchi they do not have cartilage.

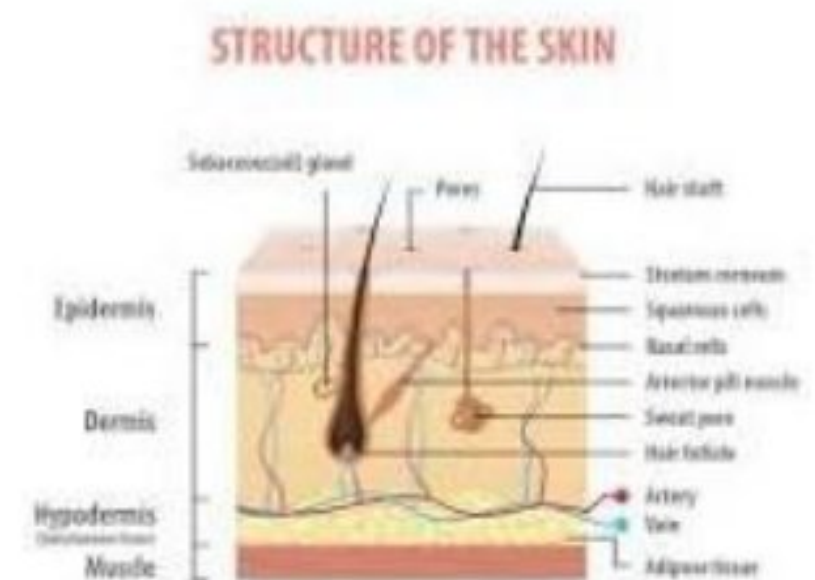
Alveolar ducts: These are terminal bronchioles divide repeatedly to form minute passages. **Alveoli:** Alveoli are the final terminations of each bronchi. **Lungs:** Lungs are the principle organs concerned with respiratory process they are two in number they are spongy organs. They lie in thoracic cavity. (6m)



13. Describe the structure of skin, draw diagram(3+3=6)

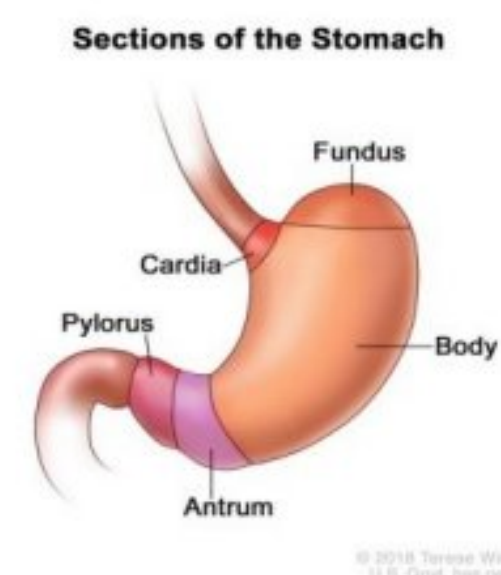
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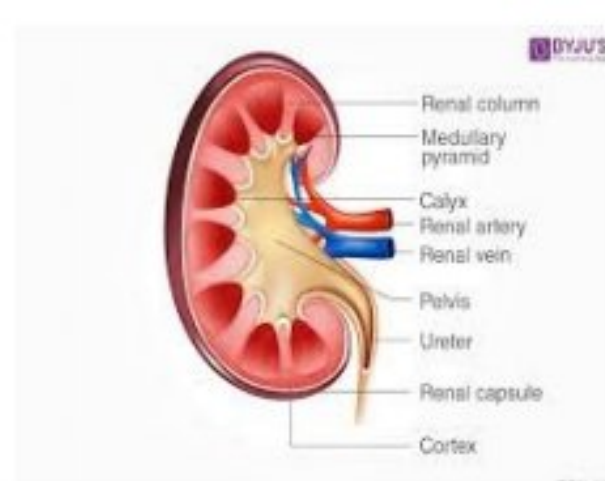


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15. Write the structure of kidney.(6m)



A:- Kidney is the main organ of urinary system it is bean shaped organ lying on posterior wall of upper abdomen one on each side of vertebral column. They lie at the level of 12th thoracic to 3rd lumbar vertebrae. Right kidney is located slightly lower than left kidney dimensions of each kidney are 11X 5 X 3cm. Each kidney weighs 150 gms approximately on the upper pole of each kidney and adrenal gland is present. Each kidney is embedded by fat called Perirenal fat each kidney

is convex on its outer border and concave in the centre of its inner border at this point blood vessels, nerves and ureter enter and leave kidney. This point of kidney is called hilus. (3m)

Longitudinal section of kidney: kidney surrounded by a fibrous capsule. It can be stripped off easily portion inside this fibrous capsule can be divided into cortex and medulla. Cortex is the outer reddish brown colour portion and medulla is the inner lighter area. Medulla is subdivided into 10 to 15 conical areas called

renal pyramids. Pyramids have their broad base towards cortex and apex projecting into lumen of minor calyx. **Microscopic structure of kidney:** Kidney is consists of number of minute units called nephrons. They are basic structural and functional units of kidney. They are about 1 mill nephrons in each kidney(3m)

16. Define B.P. and explain the method of measurement.(6m)

A:-“ Lateral pressure exerted by flowing blood on walls of blood vessels is called as blood pressure”.(2m). There are two methods for measurement of arterial blood pressure. They are: 1. Direct method of canulation of artery and 2. Indirect method not requiring any surgical procedure.

Indirect method: It is universally used method for measuring and recording. Systolic and diastolic B.P. Indirect method involves measurement of pressure of brachial artery using sphygmo manometer. There are three methods in indirect procedure. a). Oscillatory method b). Palpatory method c). Auscultatory method.

Oscillatory method: In this method, a pressure cuff is wrapped over the brachial artery. Oscillations produced in spring gauge or mercury manometer are observed.

Palpatory method: In this method. Instrument is kept at the level of heart. Cuff is tied round the upper arm. Pressure is raised to 200mm of Hg and gradually released. Pressure, when pulse appears at the wrist is noted. It is systolic B.P. Diastolic B.P. can not be measured by this method.

Auscultatory method: In this method also, instrument is kept at the level of heart. Cuff is tied round upper arm and pressure raised above 200mm of Hg and gradually released. Chest piece of steth is placed over the brachial artery just below the cuff. Pressure, when clear tapping sound appears is noted. It is systolic. B.P. Pressure, when loud sound gets muffled and rapidly begins to fade is also noted. It is diastolic B.P.(4m)

17. What are different endocrine glands of body? Write the anatomy of pituitary.(6m)

A:- The different endocrine glands of body are 1). Pituitary gland (master gland) 2). Thyroid gland 3). Parathyroid gland 4). Adrenal glands 5). Pancreas 6). Testes 7). Ovaries 8). Placenta.(2m)

Anatomy of pituitary:- It is called the master gland of the body. It is substituted by the term hypophysis. it is reddish grey coloured and small oval shaped structure. It is located at the base of the brain in the sellaturcica of sphenoid bone. A stalk attaches hypophysis to the floor of third ventricle. Average weight is 0.5 to 0.6g. Its dimensions are 10mm., (antero posteriorly), 6mm.(dorsoventrally), 13mm (laterally)

Anatomically it has two lobes – anterior lobe and posterior lobe.

Histology shows 6parts :- 1). Pars distalis 2). Pars tuberalis 3). Pars intermedia 4). Pars nervosa 5). Median eminence of tubercinerium 6). Infundibulum or pituitary stalk (4m)

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